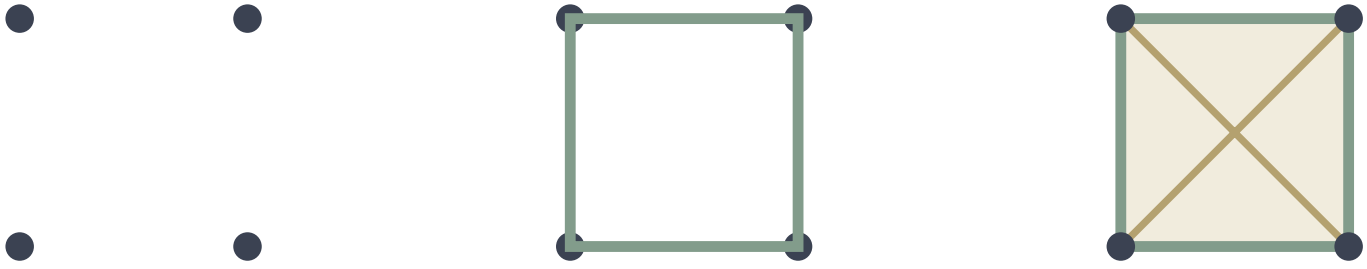
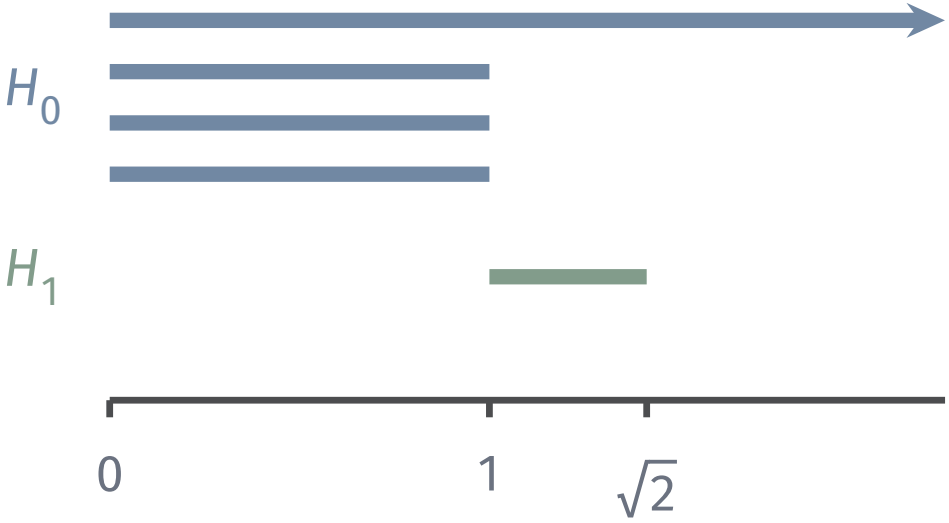


$r < 1$

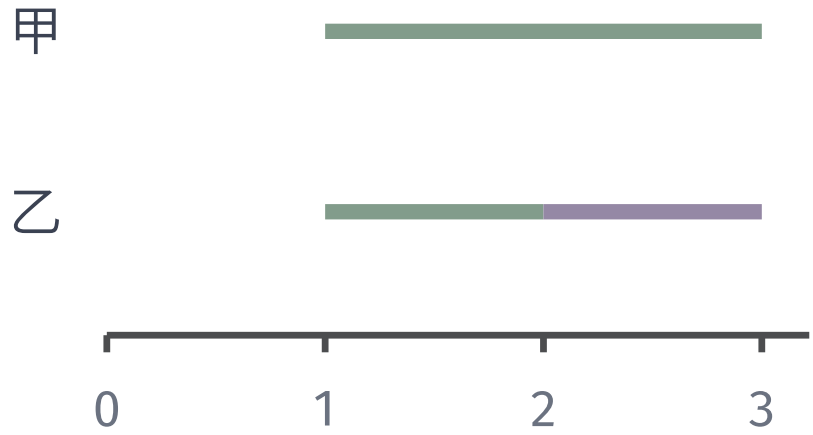
$1 \le r < \sqrt{2}$

$r \ge \sqrt{2}$

(a)



(b)



(c)